

Logic and proofs

Logic is the science dealing with the methods of reasoning. Reasoning plays a very important role in every area of knowledge, particularly mathematics.

Or

Logic is the study and analysis of the nature of the valid argument.

Propositions Consider the following sentences:

(1) Bangalore is in Karnataka.

(2) Three is a prime number.

(3) Seven is divisible by 3.

(4) Every rectangle is a square

Each of the above sentences is a statement (declaration) which can be decisively said to either true or false. In fact, sentences 1 and 2 are true statements whereas sentences 3 and 4 are false statements.

Sentences of this type are called propositions. A proposition is formally defined as follows:

A proposition is a statement (declaration) which, in a given context, can be said to be either true or false, but not both.

It should be noted that not all sentences are propositions.

For example, consider the following sentences:

1. Take a triangle ABC.

2. What is your name?

3. $xy = yx$.

4. $x+y = 10$.

We note that the sentences 1 and 2 are not at all statements; as such, it is not a proposition. The 3 and 4 sentences are statements; but we cannot decisively say whether it is true or unless we know what x and y are. As such, it is also not a proposition. However, if the context tells that x and y are real numbers, then it becomes a true proposition.

Propositional variables (or statement variables), that is, variables that represent propositions, just as letters are used to denote numerical variables. Propositions are usually represented by small letters such as p , q , r , s ...

The truth or falsity of a proposition is called its **truth value**. If a proposition is true, we will indicate truth value by the symbol T or 1 and if it is false by the symbol F or 0 .

Propositional logic: The area of logic that deals with propositions is called the propositional calculus or propositional logic.

For example, if we represent the proposition "Three is a prime number" by p , then the truth value of p is T or 1 . Similarly, if we represent the proposition "Every rectangle is a square" by q then the truth value of q is F or 0 .

Logical Connectives (logical operators) and Truth Tables

New propositions are obtained by starting with given propositions with the aid of words or phrases like 'or', 'not', 'and', 'if... then', and 'if and only if'. Such words or phrases are called logical connectives.

The new propositions obtained by the use of connectives are called **compound propositions**.

The original propositions from which a compound proposition is obtained are called the **components or the primitives of the compound proposition**.

Propositions which do not contain any logical connective are called **simple propositions**.

Negation

A proposition obtained by inserting the word 'not' at an appropriate place in a given proposition is called the negation of the given proposition. The negation of a proposition p is denoted by $\neg p$ (or $\sim p$ read 'not p '), the symbol $\neg$ denoting the word not.

It is obvious that, for any proposition p , if p is true, then $\neg p$ is false, and if p is false, then $\neg p$ is true. This observation can be recorded in the form of a Truth Table as given below:

p	$\neg p$
T	F
F	T

The first column of the above table gives the possible truth values of p and the second column gives the corresponding truth values of $\neg p$.

For example, let the proposition "3 is a prime number" be denoted by p ; that is, p : 3 is a prime number. Then the negation of p is "3 is not a prime number"; that is, $\neg p$: 3 is not a prime number.

Disjunction

A compound proposition obtained by combining two given propositions by inserting the word 'or' in between them is called the disjunction of the given propositions.

The disjunction of two propositions p and q is denoted by $p \vee q$ (read " p or q "), the symbol $\vee$ denoting the word or.

The disjunction $p \vee q$ is false only when p is false and q is false; in all other cases it is true.

This rule can be expressed more explicitly in the form of a truth table given below

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

In the above table, the last column gives the truth values of $p \vee q$ for all possible combinations of the truth values of p and q given in the first two columns.

As an illustration, let us consider the following propositions:

p : $\sqrt{2}$ is an irrational number.

r : All triangles are equilateral.

q : 9 is a prime number.

s: $2+5=7$.

Then,

$p \vee q$: $\sqrt{2}$ is an irrational number or 9 is a prime number.

$q \vee r$: 9 is a prime number or all triangles are equilateral.

$r \vee s$: All triangles are equilateral or $2+5=7$.

$s \vee p$: $2+5=7$ or $\sqrt{2}$ is an irrational number.

We note that p and s are true, and q and r are false.

Hence $p \vee q$ is true, $q \vee r$ is false, $r \vee s$ is true, and $s \vee p$ is true.

Conjunction

A compound proposition obtained by combining two given propositions by inserting the word "and" in between them is called the conjunction of the given propositions. The conjunction of two propositions p and q is denoted by $p \wedge q$ (read "p and q"), the symbol $\wedge$ denoting the word and.

The conjunction $p \wedge q$ is true only when p is true and q is true; in all other cases it is false .

This rule can be expressed more explicitly in the form of a truth table as given below:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

In the above table, the last column gives the truth values of $p \wedge q$ for all possible combinations of the truth values of p and q given in the first two columns.

As an illustration, let us consider the following propositions.

p: $\sqrt{2}$ is an irrational number.

q: 9 is a prime number.

r: All triangles are equilateral.

s: $2+5=7$.

Here, p and s are true propositions and q and r are false propositions.

Then $p \wedge q$: $\sqrt{2}$ is an irrational number and 9 is a prime number.

$q \wedge r$: 9 is a prime number and all triangles are equilateral.

$r \wedge s$: All triangles are equilateral and $2+5=7$

$s \wedge p$: $2+5=7$ and $\sqrt{2}$ is an irrational number.

Since p is true and q is false, $p \wedge q$ is false. Since both q and r are false, $q \wedge r$ is false. Since r is false and s is true, $r \wedge s$ is false. Since both s and p are true, $s \wedge p$ is true.

Conditional

A two proposition p and q connected by the words 'if p then q' than it is called a conditional. And it is denoted by $p \rightarrow q$.

The conditional $p \rightarrow q$ is false only when p is true and q is false; in all other cases it is true.

In other words, the truth value of the conditional $p \rightarrow q$ is F only when the truth value of p

is T and the truth value of q is F; in all other cases, the truth value of $p \rightarrow q$ is T.
 It is important to note that the conditional $q \rightarrow p$ is not the same as the conditional $p \rightarrow q$.
 This rule can be expressed more explicitly in the form of a truth table as given below

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

As an illustration, let us consider the following propositions,

p: 2 is a prime number.

q: 3 is a prime number.

r: 6 is a perfect square.

s: 9 is a multiple of 6.

Now,

$p \rightarrow q$: If 2 is a prime number, then 3 is a prime number.

$p \rightarrow r$: If 2 is a prime number, then 6 is a perfect square.

$r \rightarrow p$: If 6 is a perfect square, then 2 is a prime number.

$r \rightarrow s$: If 6 is a perfect square, then 9 is a multiple of 6.

Here, p and q are true propositions, and r and s are false propositions.

Hence $p \rightarrow q$ is true, $p \rightarrow r$ is false, $r \rightarrow p$ and $r \rightarrow s$ is true.

Biconditional:

A two proposition p and q connected by the words ‘p iff q’ than it is called a biconditional. And it is denoted by $p \leftrightarrow q$.

The compound proposition is true if p and q are same values otherwise false.

This rule can be expressed more explicitly in the form of a truth table as given below:

p	q	$P \leftrightarrow q.$
T	T	T
T	F	F
F	T	F
F	F	T

From the above table, it is clear that $p \leftrightarrow q$ is true only when both p and q have the same truth values.

For example, let

p: 2 is a prime number.

q: 3 is a prime number.

r: 6 is a perfect square.

s: 9 is a multiple of 6.

Then the biconditionals $p \leftrightarrow q$ and $r \leftrightarrow s$ are true and the biconditionals $p \leftrightarrow r$, $p \leftrightarrow s$

$q \leftrightarrow r$ and $q \leftrightarrow s$ are false.

Exclusive Disjunction

The compound proposition "p or q" to be true only when either p is true or q is true but not both. In other word Exclusive OR or exclusive disjunction is a logical operation that is true if and only if its arguments differ (one is true, the other is false).

The exclusive or is denoted by the symbol $\vee$ (or $\oplus$).

The compound proposition $p \vee q$ (read as either p or q but not both) is called the exclusive disjunction of the propositions p and q.

Its truth table is as given below.

p	q	$p \vee q$
T	T	F
T	F	T
F	T	T
F	F	F

For example, let

p: $\sqrt{2}$ is an irrational number.

q: $2+3=5$.

r: 9 is a prime number.

s: All triangles are isosceles.

Then

$p \vee q$: Either $\sqrt{2}$ is an irrational number or $2+3=5$, but not both.

$q \vee r$: Either $2+3=5$ or 9 is a prime number, but not both.

$r \vee s$: Either 9 is a prime number or all triangles are isosceles, but not both.

$s \vee p$: Either all triangles are isosceles or $\sqrt{2}$ is an irrational number, but not both.

Since p and q are true and r and s are false, we note that $p \vee q$ is false, $q \vee r$ is true, $r \vee s$ is false, and $s \vee p$ is true.

Logical Operators and symbols:

Operators	Representation	Meaning
Negation	$\neg p$	"not p"
Disjunction	$p \vee q$	"p or q"
Conjunction	$p \wedge q$	"p and q"
Conditional	$p \rightarrow q$	if p then q"
Biconditional	$P \leftrightarrow q$	"p if and only if q"
Exclusive disjunction	$p \vee q$	"either p or q, but not both"

Example .

Write the following compound propositions symbolically.

- (a) 5 is a prime or a composite number
- (b) Nirmala is practical and quiet.
- (c) He studies hard and does not understand logic
- (d) If all odd integers are prime and 4 divides 6, then it will not rains.

Solution

(a) Let p denote "5 is prime" and q be denoted as "5 is a composite number". Then the given proposition is $p \vee q$

(b) Consider p : Nirmala is practical and q : Nirmala is quiet. Then the given proposition is $p \wedge q$

(c) Let p : He studies hard and q : He understands logic. $\neg q$: he does not understand logic. Then the given proposition is $p \wedge (\neg q)$

(d) Let, p : All odd integers are primes, q : 4 divides 6; r : It rains

Now $\neg r$: It will not rain.

Then the given proposition is: $(p \wedge q) \rightarrow (\neg r)$

Example:

Let p and q are the propositions

p : I bought a lottery ticket this week

q : I won the million dollar jackpot on Friday

Express each of these propositions as an English sentence.

(a) $\neg p$

(b) $p \vee q$

(c) $p \rightarrow r$

(d) $p \wedge q$

(e) $p \leftrightarrow r$

(f) $\neg p \rightarrow \neg r$

(g) $\neg p \wedge \neg q$

(h) $\neg p \vee (p \wedge q)$

Solution:

(a) I didn't buy lottery ticket, this week,

(b) I bought a lottery ticket this week or I won the million dollar jackpot on Friday

(c) If I bought a lottery ticket this week then I won the million dollar jackpot on Friday.

(d) I bought a lottery ticket this week and I won the million dollar jackpot on Friday.

(e) I bought the lottery ticket this week if and only if I won the million dollar jackpot on Friday.

(f) If I don't buy the lottery ticket this week then I would not win the million dollar jackpot on Friday.

(g) I didn't buy the lottery ticket this week and I would not win the million dollar jackpot on Friday.

(h) I didn't buy the lottery ticket this week or I bought the lottery ticket this week and won the million dollar jackpot on Friday.

Precedence of logical operators:

For the logical operators, the rules are that $\neg$ has higher precedence than $\wedge$, and $\wedge$ has precedence over $\vee$, $\vee$ has precedence over $\rightarrow$, $\rightarrow$ has precedence over $\leftrightarrow$. This means that in the absence of parentheses, any $\neg$ operators are evaluated first, followed by any $\wedge$ operators, followed by any $\vee$ operators, followed by $\rightarrow$ operators, followed by $\leftrightarrow$ operators.

Operators	Precedence
$\neg$	1
$\vee$	2
$\wedge$	3
$\rightarrow$	4
$\leftrightarrow$	5

Tautology and Contradiction

Definition: A compound proposition is said to be a tautology if it is always true for all possible combinations of truth values of its components.

Definition: A compound proposition is said to be a contradiction if it is always false for all possible combinations of the truth values of its components.

Inverse, Converse and Contrapositive

Consider an implication $p \rightarrow q$ then

- (i) $(\neg p) \rightarrow (\neg q)$ is called Inverse of $p \rightarrow q$
- (ii) $q \rightarrow p$ is called Converse of $p \rightarrow q$
- (iii) $(\neg q) \rightarrow (\neg p)$ is called Contrapositive of $p \rightarrow q$

Write the inverse, converse and contrapositive for the following

(1) If two integers are equal then their squares are equal.

Sol: p : Two integers are equal.

q : The squares of two integer are equal.

Inverse: $(\neg p) \rightarrow (\neg q)$

If two integers are not equal then their square of two integers are also not equal.

Converse: $q \rightarrow p$

If two integers squares are equal then the two integers are equal.

Contrapositive : $(\neg q) \rightarrow (\neg p)$

If two integers squares are not equal. Then the two integers are not equal.

(2) If I work hard, then I get the grade

Sol. Here p : I work hard

q : I get the grade

$\therefore p \rightarrow q$

Inverse: $(\neg p) \rightarrow (\neg q)$

If I won't work hard then I won't get the grade.

Converse: $q \rightarrow p$

If I get the grade then I work hard

Contrapositive: $(\neg q) \rightarrow (\neg p)$

If I won't get the grade then I won't work hard.

(3) If 2 is not a prime number then it is an even number.

Sol. Here

p : 2 is not a prime number

q: 2 is an even number

Inverse:

If 2 is a prime number then it is not an even number

Converse:

If 2 is an even number then it is not a prime number

Contrapositive:

If 2 is not an even number then it is a prime number

Propositional equivalence or logical equivalence:

Two propositions u and v are said to be logically equivalent whenever u and v have the same truth value, or equivalently, the biconditional $u \leftrightarrow v$ is a tautology.

Then, we write $u \Leftrightarrow v$ (or $u \equiv v$) Here the symbol $\Leftrightarrow$ stands for "logically equivalent to". Logically equivalent propositions are treated as identical propositions."

Example: Let x be a specified positive integer. Consider the following

p : x is an odd integer.

q : x is not divisible by 2.

Are p and q logically equivalent?

We note that p and q have the same truth values. As such, p and q are logically equivalent;

That is, $p \Leftrightarrow q$.

Example 1: For any two propositions p, q , prove that $(p \rightarrow q) \Leftrightarrow (\neg p \vee q)$.

Truth table:

p	q	$p \rightarrow q$	$(\neg p) \vee q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

Here $(p \rightarrow q)$ and $(\neg p \vee q)$ have the same truth values for all possible truth values of p and q .

Therefore, $(p \rightarrow q) \Leftrightarrow (\neg p \vee q)$.

Example 2 : Prove that ,for any three propositions p, q, r .

$[(p \vee q) \rightarrow r] \Leftrightarrow [(p \rightarrow r) \wedge (q \rightarrow r)]$

Solution : Truth table

p	q	r	$p \vee q$	$(p \vee q) \rightarrow r$	$p \rightarrow r$	$q \rightarrow r$	$(p \rightarrow r) \wedge (q \rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F
T	F	T	T	T	T	T	T
T	F	F	T	F	F	T	F
F	T	T	T	T	T	T	T
F	T	F	T	F	T	F	F
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T

Here $(p \vee q) \rightarrow r$ and $[(p \rightarrow r) \wedge (q \rightarrow r)]$ have the same values in all possible situations.

Therefore, $[(p \vee q) \rightarrow r] \Leftrightarrow [(p \rightarrow r) \wedge (q \rightarrow r)]$.

Example 3: Show that $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent.

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

Here $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ have the same values in all possible situations.
Therefore, $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$.

The laws of logic:

Table 6 contains some important equivalence. In these equivalences, T denotes the compound proposition that is always true and F denotes the compound proposition that is always false. We also display some useful equivalence for compound propositions involving conditional statements and biconditional statements in Tables 7 and 8, respectively.

TABLE 6 Logical Equivalences.		TABLE 7 Logical Equivalences Involving Conditional Statements.	TABLE 8 Logical Equivalences Involving Biconditional Statements.
Equivalence	Name	$p \rightarrow q \equiv \neg p \vee q$ $p \rightarrow q \equiv \neg q \rightarrow \neg p$ $p \vee q \equiv \neg p \rightarrow q$ $p \wedge q \equiv \neg(p \rightarrow \neg q)$ $\neg(p \rightarrow q) \equiv p \wedge \neg q$ $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$ $(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$ $(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$ $(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$ $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$ $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$ $\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$
$p \wedge T \equiv p$ $p \vee F \equiv p$ $p \vee T \equiv T$ $p \wedge F \equiv F$	Identity laws		
$p \vee T \equiv T$ $p \wedge F \equiv F$	Domination laws		
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws		
$\neg(\neg p) \equiv p$	Double negation law		
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws		
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws		
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws		
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws		
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws		
$p \vee \neg p \equiv T$ $p \wedge \neg p \equiv F$	Negation laws		

Prove the following without using truth table

1) $p \vee q \vee (\neg p \wedge \neg q \wedge r) \Leftrightarrow (p \vee q \vee r)$

Solution:

$$\begin{aligned}
 & p \vee q \vee (\neg p \wedge \neg q \wedge r) \\
 & \Leftrightarrow (p \vee q) \vee ((\neg p \wedge \neg q) \wedge r) && \text{by using associative law} \\
 & \Leftrightarrow (p \vee q) \vee (\neg(p \vee q) \wedge r) && \text{by using de morgan's law} \\
 & \Leftrightarrow ((p \vee q) \vee \neg(p \vee q)) \wedge ((p \vee q) \vee r) && \text{by using distributive law} \\
 & \Leftrightarrow T \wedge ((p \vee q) \vee r) && \text{by using negation law} \\
 & \Leftrightarrow ((p \vee q) \vee r) && \text{by using identity law} \\
 & \Leftrightarrow (p \vee q \vee r) && \text{by using associative law}
 \end{aligned}$$

2) $(p \rightarrow q) \wedge [\neg q \wedge (r \vee \neg q)] \Leftrightarrow \neg(q \vee p)$

Solution:

$$\begin{aligned}
 & (p \rightarrow q) \wedge [\neg q \wedge (r \vee \neg q)] \\
 & \Leftrightarrow (p \rightarrow q) \wedge (\neg q \wedge (r \vee \neg q)) && \text{by using commutative law} \\
 & \Leftrightarrow (p \rightarrow q) \wedge (\neg q) && \text{by an absorption law} \\
 & \Leftrightarrow \neg[(p \rightarrow q) \rightarrow \neg(\neg q)]. && \text{Because } (u \wedge v) \Leftrightarrow \neg(u \rightarrow \neg v) \\
 & \Leftrightarrow \neg[(p \rightarrow q) \rightarrow q] && \text{by using double negation law} \\
 & \Leftrightarrow \neg[\neg(p \rightarrow q) \vee q] && \text{because } u \rightarrow v \Leftrightarrow \neg u \vee v \\
 & \Leftrightarrow \neg[(p \wedge \neg q) \vee q] && \text{because } \neg(u \rightarrow v) \Leftrightarrow u \wedge \neg v \\
 & \Leftrightarrow \neg[q \vee (p \wedge \neg q)], && \text{by commutative law} \\
 & \Leftrightarrow \neg[(q \vee p) \wedge (q \vee \neg q)] && \text{by distribution law} \\
 & \Leftrightarrow \neg((q \vee p) \wedge T) && \text{by negation law}
 \end{aligned}$$

$$\Leftrightarrow \neg (q \vee p) \quad \text{by identity law.}$$

This proves the required result.

$$3) \neg (\neg ((p \vee q) \wedge r) \vee \neg q) \Leftrightarrow (q \wedge r)$$

Solution:

$$\begin{aligned} \neg (\neg ((p \vee q) \wedge r) \vee \neg q) & \\ \Leftrightarrow \neg (\neg ((p \vee q) \wedge r) \wedge q) & \quad \text{by De Morgan's law} \\ \Leftrightarrow ((p \vee q) \wedge r) \wedge q & \quad \text{by double negation law} \\ \Leftrightarrow (p \vee q) \wedge (r \wedge q) & \quad \text{by associative law} \\ \Leftrightarrow (p \vee q) \wedge (q \wedge r) & \quad \text{by commutative law} \\ \Leftrightarrow ((p \vee q) \wedge q) \wedge r & \quad \text{by associative law} \\ \Leftrightarrow (q \wedge r) & \quad \text{by absorption law} \end{aligned}$$

Predicates and Quantifiers:

Predicates:

A predicate is a sentence that contains a finite number of variables and becomes a statement (proposition) when specific values are substituted for the variables.

Or

An open sentence is a statement which contains an unspecified symbol called a variable. It is called a predicate and it is denoted by $p(x)$.

Replacement set: A set is called a replacement for the open sentence $p(x)$: is $\forall a \in S$, $p(a)$ is a proposition.

The domain of a predicate variable is the set of all values that may be substituted in place of variables.

Examples:

$p(x)$: "x is greater than 3",

$p(x)$: "x = y + 3"

The statement "x is greater than 3" has two parts. The first, the variable x, is the subject of the statement. The second part is the predicate.

"is greater than 3" refers to the property that the subject of the statement can have.

We can denote the statement "x is greater than 3" by $p(x)$, where p denotes the predicate "is greater than 3" and x is a variable.

The statement $p(x)$ is also said to be the value of the propositional function p at x.

Once a value has been assigned to the variable x, the statement $p(x)$ becomes a proposition and has a truth value.

Example 1: Let $P(x)$ denote the statement " $x > 3$." What are the truth values of $P(4)$ and $P(2)$?

Solution: We obtain the statement $P(4)$ by setting $x = 4$ in the statement " $x > 3$." Hence, $P(4)$, which is the statement " $4 > 3$," is true. However, $P(2)$, which is the statement " $2 > 3$," is false. We can also have statements that involve more than one variable. For instance, consider the statement " $x = y + 3$." We can denote this statement by $Q(x, y)$, where x and y are variables and Q is the predicate. When values are assigned to the variables x and y, the statement $Q(x, y)$ has a truth value.

Example 2: Let $Q(x, y)$ denote the statement " $x = y + 3$." What are the truth values of the propositions $Q(1, 2)$ and $Q(3, 0)$?

Solution: To obtain $Q(1, 2)$, set $x = 1$ and $y = 2$ in the statement $Q(x, y)$. Hence, $Q(1, 2)$ is the statement " $1 = 2 + 3$," which is false. The statement $Q(3, 0)$ is the proposition " $3 = 0 + 3$," which is true.

Similarly, we can let $R(x, y, z)$ denote the statement " $x + y = z$." When values are assigned to the variables x, y, and z, this statement has a truth value.

Example 3: What are the truth values of the propositions $R(1, 2, 3)$ and $R(0, 0, 1)$?

Solution: The proposition $R(1, 2, 3)$ is obtained by setting $x = 1$, $y = 2$, and $z = 3$ in the statement $R(x, y, z)$. We see that $R(1, 2, 3)$ is the statement “ $1 + 2 = 3$,” which is true. Also note that $R(0, 0, 1)$, which is the statement “ $0 + 0 = 1$,” is false.

In general, a statement involving the n variables $x_1, x_2, \dots, x_n$ can be denoted by $P(x_1, x_2, \dots, x_n)$. A statement of the form $P(x_1, x_2, \dots, x_n)$ is the value of the propositional function P at the n -tuple $(x_1, x_2, \dots, x_n)$, and P is also called an n -place predicate or an n -ary predicate.

Quantifiers:

Any proposition contain the idea of quantity like all, for every, there exist such words are called **quantifiers**.

Quantification expresses the extent to which a predicate is true over a range of elements. Or

A quantifiers tells about what quantity of element make the predicate true.

In these propositions, the words "all", "every", "some", "There exists" are associated with the idea of a quantity. Such words are called **quantifiers**.

Consider the following propositions

- (1) All square rectangles.
- (2) For every integer x , x^2 non-negative integer.
- (3) Some determinants are equal to zero.
- (4) There exists a real number whose square is equal to itself.

The area of logic that deals with predicates and quantifiers is called the **predicate calculus**.

Many mathematical statements assert that a property is Assessment true for all values of a variable in a particular domain, called the **domain of discourse (or the universe of discourse)**, and often just referred to as the domain.

The quantifiers classified in to two types:

1. The universal quantifiers: (for all, for every)

Many mathematical statements assert that a property is true all values of a variable in a particular domain, called the universe of discourse (or domain of discourse), and often just referred to as the domain, such a statement is expressed using universal quantification. It is denoted by " $\forall$ " (for all, for every)

The universal quantification of $P(x)$ is the statement
“ $P(x)$ for all values of x in the domain”

The notation $\forall x P(x)$ denotes the universal quantification of $P(x)$. Here $\forall$ is called the universal quantifier. We read $\forall x P(x)$ as “for all $x P(x)$ ” or “for every $x P(x)$.” An element for which $P(x)$ is false is called a counterexample to $\forall x P(x)$.

Besides “for all” and “for every,” universal quantification can be expressed in many other ways, including “all of,” “for each,” “given any,” “for arbitrary,” “for each,” and “for any.”

Example 1: Let $P(x)$ be the statement “ $x + 1 > x$.” What is the truth value of the quantification $\forall x P(x)$, where the domain consists of all real numbers?

Solution: Because $P(x)$ is true for all real numbers x , the quantification $\forall x P(x)$ is true.

Example 2: Let $Q(x)$ be the statement “ $x < 2$.” What is the truth value of the quantification $\forall x Q(x)$, where the domain consists of all real numbers?

Solution: $Q(x)$ is not true for every real number x , because, for instance, $Q(3)$ is false. That is, $x = 3$ is a counterexample for the statement $\forall x Q(x)$. Thus, $\forall x Q(x)$ is false.

Example 3:

Suppose that $P(x)$ is “ $x^2 > 0$.” To show that the statement $\forall x P(x)$ is false where the universe of discourse consists of all integers, we give a counterexample. We see that $x = 0$ is a counterexample because $x^2 = 0$ when $x = 0$, so that x^2 is not greater than 0 when $x = 0$.

2. The existential quantifiers: (for some, there exists, at least one)

Many mathematical statements assert that there is an element with certain property. Such statements are expressed using existential quantification. With existential quantification, we form a proposition that is true if and only if $p(x)$ is true for at least one value of x in the domain. It is denoted by " $\exists$ ".

The existential quantification of $P(x)$ is the proposition

"There exists an element x in the domain such that $P(x)$."

We use the notation $\exists x P(x)$ for the existential quantification of $P(x)$. Here $\exists$ is called the existential quantifier.

Besides the phrase "there exists," we can also express existential quantification in many other ways, such as by using the words "for some," "for at least one," or "there is."

The existential quantification $\exists x P(x)$ is read as

"There is an x such that $P(x)$,"

"There is at least one x such that $P(x)$,"

or

"For some x $P(x)$."

Example 1: Let $P(x)$ denote the statement " $x > 3$." What is the truth value of the quantification $\exists x P(x)$, where the domain consists of all real numbers?

Solution: Because " $x > 3$ " is sometimes true for instance, when $x = 4$ the existential quantification of $P(x)$, which is $\exists x P(x)$, is true.

Example 2: Let $Q(x)$ denote the statement " $x = x + 1$." What is the truth value of the quantification $\exists x Q(x)$, where the domain consists of all real numbers?

Solution: Because $Q(x)$ is false for every real number x , the existential quantification of $Q(x)$, which is $\exists x Q(x)$, is false.

Quantifiers with finite domain

When the domain of a quantifier is finite, that is, when all its elements can be listed, quantified statements can be expressed using propositional logic.

In particular, when the elements of the domain are $x_1, x_2, \dots, x_n$, where n is a positive integer, the universal quantification $\forall x P(x)$ is the same as the conjunction

$P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$,

because this conjunction is true if and only if $P(x_1), P(x_2), \dots, P(x_n)$ are all true.

Example 1: What is the truth value of $\forall x P(x)$, where $P(x)$ is the statement " $x^2 < 10$ " and the domain consists of the positive integers not exceeding 4?

Solution: The statement $\forall x P(x)$ is the same as the conjunction

$P(1) \wedge P(2) \wedge P(3) \wedge P(4)$,

because the domain consists of the integers 1, 2, 3, and 4. Because $P(4)$, which is the statement

" $4^2 < 10$," is false, it follows that $\forall x P(x)$ is false.

Similarly, when the elements of the domain are $x_1, x_2, \dots, x_n$, where n is a positive integer, the

existential quantification $\exists x P(x)$ is the same as the disjunction $P(x_1) \vee P(x_2) \vee \dots \vee P(x_n)$,

because this disjunction is true if and only if at least one of $P(x_1), P(x_2), \dots, P(x_n)$ is true.

Example 2: What is the truth value of $\exists x P(x)$, where $P(x)$ is the statement " $x^2 > 10$ " and the domain consists of the positive integers not exceeding 4?

Solution: Because the domain is $\{1, 2, 3, 4\}$, the proposition $\exists x P(x)$ is the same as the disjunction

$P(1) \vee P(2) \vee P(3) \vee P(4)$.

Because $P(4)$, which is the statement " $4^2 > 10$," is true, it follows that $\exists x P(x)$ is true.

Quantifiers with Restricted Domains

Example: What do the statements $\forall x < 0 (x^2 > 0)$, $\forall y \neq 0 (y^3 \neq 0)$, and $\exists z > 0 (z^2 = 2)$ mean, where the domain in each case consists of the real numbers?

Solution: The statement $\forall x < 0 (x^2 > 0)$ states that for every real number x with $x < 0$, $x^2 > 0$.

That is, it states "The squares of a negative real number are positive."

The statement $\forall y \neq 0 (y^3 \neq 0)$ states that for every real number y with $y \neq 0$, we have $y^3 \neq 0$. That is, it states "The cube of every nonzero real number is nonzero."

Finally, the statement $\exists z > 0 (z^2 = 2)$ states that there exists a real number z with $z > 0$ such that $z^2 = 2$. That is, it states "There is a positive square root of 2."

Expressing quantifications in English

Example 1:

Let $P(x)$ be the statement "x has visited Bangalore" where the domain consists of the students in your college. Express $\forall x P(x)$ and $\exists x P(x)$ in English.

Solution:

The quantification $\forall x P(x)$ denotes the proposition

For every x in the domain of the students in your college, x has visited Bangalore. Or all students in your college have visited Bangalore.

The quantification $\exists x P(x)$ denotes the proposition

For some x in the domain of the students in your college, x has visited Bangalore. Or some students in your college have visited Bangalore.

Example 2:

Let $P(x)$ be the statement "x spends more than five hours every weekday in class" where the domain consists of the students in your college. Express $\forall x P(x)$ and $\exists x P(x)$ in English.

Solution:

The quantification $\forall x P(x)$ denotes the proposition

For every x in the domain of the students in your college, x spends more than five hours every weekday in class.

or All students in your college spends more than five hours every weekday in class. The quantification

$\exists x P(x)$ denotes the proposition

For some x in the domain of the students in your college, x spends more than five hours every weekday in class. Or Some students in your college spends more than five hours every weekday in class.

Negating Quantified Expressions

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x).$$

Example 1:

"Every student in your class has taken a course in calculus." This statement is a universal quantification, namely, $\forall x P(x)$, where $P(x)$: "x has taken a course in calculus" and the domain consists of students in your class. The negation of this statement is $\exists x \neg P(x)$.

$\neg P(x)$: "x has not taken a course in calculus"

"There is a student in your class who has not taken a course in calculus."

Example 2:

"There is a student in this class who has taken a course in calculus." This is the existential quantification $\exists x Q(x)$, where $Q(x)$: "x has taken a course in calculus" and the domain consists of students in your class. The negation of this statement is $\forall x \neg Q(x)$.

$\neg Q(x)$: "x has not taken a course in calculus"

That is "Every student in this class has not taken a course in calculus".

Example 3:

What is the negation of the statement "All birds can fly"?

Solution:

"All birds can fly"

This statement is a universal quantification, namely, $\forall x P(x)$, where $P(x)$: "bird x can fly"

and Domain: All birds

The negation of this statement is $\exists x \neg Q(x)$.

$\neg Q(x)$: "bird x cannot fly"

That is "There are some birds who can't fly".

Example 4:

What is the negation of the statement "Most bikes are expensive"?

Solution:

"Most bikes are expensive"

This statement is a universal quantification, namely, $\exists x P(x)$, where $P(x)$: "bike x is expensive"

and Domain: All bikes

The negation of this statement is $\forall x \neg Q(x)$.

$\neg Q(x)$: "bike x is not expensive"

That is "All bikes are not expensive".

Logical Equivalences Involving Quantifiers

Two logical statements involving predicates and quantifiers are said to be equivalent if and only if they have same truth value in all the possible cases.

There are two important logical equivalences involving quantifiers,

$$\forall x (P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x)$$

$$\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$$

$$1) \forall x (P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x)$$

So, suppose that $\forall x (P(x) \wedge Q(x))$ is true.

This means that if a is in the domain, then $P(a) \wedge Q(a)$ is true. Hence, $P(a)$ is true and $Q(a)$ is true.

Because $P(a)$ is true and $Q(a)$ is true for every element a in the domain, we can conclude that

$\forall x P(x)$ and $\forall x Q(x)$ are both true.

This means that $\forall x P(x) \wedge \forall x Q(x)$ is true. Suppose that

$\forall x P(x) \wedge \forall x Q(x)$ is true.

It follows that $\forall x P(x)$ is true and $\forall x Q(x)$ is true. Hence, if a is in the domain, then $P(a)$ is true and $Q(a)$ is true.

It follows that for all a , $P(a) \wedge Q(a)$ is true. It follows that

$\forall x (P(x) \wedge Q(x))$ is true.

Hence, $\forall x (P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x)$.

Similarly,

$$\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x).$$

Nested quantifiers:

Two quantifiers are nested if one is within the scope of the other.

Example: $\forall x \exists y P(x, y)$.

The order of quantifiers Example 1:

Let $P(x, y)$ is $(xy = yx)$, What are the truth values of the quantifications $\forall x \forall y P(x, y)$ and $\forall y \forall x P(x, y)$ where the domain for all variables consists of all integers ?

Solution:

The quantification $\forall x \forall y P(x, y)$ denotes the proposition "For all real numbers x , for all real numbers y , $xy = yx$ ". This is true.

The quantification $\forall y \forall x P(x, y)$ denotes the proposition

"For all real numbers y , for all real numbers x , $xy = yx$ ".

This is true.

Hence, The order of nested universal quantifiers in a statement without other quantifiers can be changed without changing the meaning of the quantified statement.

That is $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$.

Example 2:

Let $P(x, y)$ is $(xy = 6)$. What are the truth values of the quantifications $\exists x \exists y P(x, y)$ and $\exists y \exists x P(x, y)$ where the domain for all variables consists of all integers ?

Solution:

The quantification $\exists x \exists y P(x, y)$ denotes the proposition

"There is an integer x for which there is an integer y that $xy = 6$ ".

This is true.

The quantification $\exists y \exists x P(x, y)$ denotes the proposition

"There is an integer y for which there is an integer x that $xy = 6$ ". This is true.

Hence, The order of nested existential quantifiers in a statement without other quantifiers can be changed without changing the meaning of the quantified statement.

That is $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$.

Example 3:

Let $P(x, y)$ be the statement " $x + y = y + x$." What are the truth values of the quantifications $\forall x \forall y P(x, y)$ and $\forall y \forall x P(x, y)$, where the domain for all variables consists of all real numbers?

Solution:

The quantification $\forall x \forall y P(x, y)$ denotes the proposition

"For all real numbers x , for all real numbers y , $x + y = y + x$."

Because $P(x, y)$ is true for all real numbers x and y , the proposition $\forall x \forall y P(x, y)$ is true. Note that the statement $\forall y \forall x P(x, y)$ says "For all real numbers y , for all real numbers x , $x + y = y + x$."

This has the same meaning as the statement "For all real numbers x , for all real numbers y , $x + y = y + x$."

That is, $\forall x \forall y P(x, y)$ and $\forall y \forall x P(x, y)$ have the same meaning, and both are true.

Example 4:

Let $Q(x, y)$ denote " $x + y = 0$." What are the truth values of the quantifications $\forall x \forall y Q(x, y)$, $\exists y \forall x Q(x, y)$, $\forall x \exists y Q(x, y)$ and $\exists x \exists y Q(x, y)$, where the domain for all variables consists of all real numbers?

Solution:

The quantification $\forall x \forall y Q(x, y)$ denotes the proposition "For all real numbers x , for all real numbers y , $x + y = 0$."

For all real numbers x and y , the proposition $x + y = 0$ is not true.

So $\forall x \forall y Q(x, y)$ is false.

The quantification $\exists y \forall x Q(x, y)$ denotes the proposition

"There is a real number y such that for every real number x , $x + y = 0$."

No matter what value of y is chosen, there is only one value of x for which $x + y = 0$. Because there is no real number y such that $x + y = 0$ for all real numbers x , the statement $\exists y \forall x Q(x, y)$ is false.

The quantification $\forall x \exists y Q(x, y)$ denotes the proposition

"For every real number x there is a real number y such that $x + y = 0$."

Given a real number x , there is a real number y such that $x + y = 0$; namely, $y = -x$. Hence, the statement $\forall x \exists y Q(x, y)$ is true.

The quantification $\exists x \exists y Q(x, y)$ denotes the proposition

"There is an integer x for which there is an integer y that, $x + y = 0$ ".

There is a real number x , there is a real number y such that $x + y = 0$; namely, $x = -y$. Hence, the statement $\exists x \exists y Q(x, y)$ is true.

Example 5:

Let $Q(x, y, z)$ be the statement " $x + y = z$." What are the truth values of the statements $\forall x \forall y \exists z Q(x, y, z)$ and $\exists z \forall x \forall y Q(x, y, z)$, where the domain of all variables consists of all real numbers?

Solution:

Suppose that x and y are assigned values. Then, there exists a real number z such that $x + y = z$.

Consequently, the quantification $\forall x \forall y \exists z Q(x, y, z)$, which is the statement

"For all real numbers x and for all real numbers y there is a real number z such that $x + y = z$," is true, because for all value of x and y there is some real number that satisfies the equation $x + y = z$.

The quantification $\exists z \forall x \forall y Q(x, y, z)$, which is the statement

"There is a real number z such that for all real numbers x and for all real numbers y it is true that $x + y = z$," is false, because there is no value of z that satisfies the equation $x + y = z$ for all values of x and y .

Methods of proof:

The process of establishing that the conditional is true by using the rules/laws of logic and other known facts constitutes a proof of the conditional.

The process of establishing that a proposition is false constitutes a disproof.

Direct Proof

The direct method of proving a conditional $p \rightarrow q$ has the following lines of argument

1. Hypothesis: First assume that p is true.
2. Analysis: Starting with the hypothesis and employing the rules/laws of logic and other known facts, infer that q is true.
3. Conclusion: $p \rightarrow q$ is true.

Example 1:

Give a direct proof of the statement:

"The square of an odd integer is an odd integer".

Solution:

Here, the conditional to be proved is:

"If n is an odd integer, then n^2 is an odd integer."

Here,

p : n is an odd integer.

q : n^2 is an odd integer.

Assume that p is true that is n is an odd integer.

Then, $n = 2k + 1$ for some integer k .

Consequently,

$$\begin{aligned} n^2 &= (2k+1)^2 = 4k^2 + 4k + 1. \\ &= 2(2k^2 + 2k) + 1 \end{aligned}$$

We know that any number multiplied by a 2 must be even. We also know that if we add 1 to any even number, it becomes odd.

That gives q is true.

Hence $p \rightarrow q$ is true.

The given statement is thus proved by a direct proof.

Example 2

Give a direct proof of the statement:

For all integers k and l , "if k and l are both odd, then $k+l$ is even and kl is odd"

Solution:

Here, the conditional to be proved is:

"if k and l are both odd, then $k+l$ is even and kl is odd"

Here,

p: k and l are both odd.

q: k+l is even and kl is odd.

Assume that p is true

Take any two integers k and l, and assume that both of these are odd.

Then for some integers m and n

$$k = 2m+1.$$

$$l = 2n+1$$

Therefore,

$$k+l = (2m+1)+(2n+1)=2(m+n+1)$$

$$\begin{aligned} kl &= (2m+1)(2n+1)=4mn + 2m + 2n + 1. \\ &= 2(2mn+m+n)+1 \end{aligned}$$

We know that any number multiplied by an 2 must be even. We also know that if we add 1 to any even number, it becomes odd.

We observe that k+l is multiple of 2 and kl is not multiple of 2.

Therefore, k+l is an even integer and kl is an odd integer.

That gives q is true.

Hence $p \rightarrow q$ is true.

The given statement is thus proved by a direct proof.

Example 3:

Give a direct proof of the statement:

If a and b are consecutive integers, then sum of a+b must be an odd number.

Solution:

Here, the conditional to be proved is:

If a and b are consecutive integers, then sum of a+b must be an odd number.

Here,

p: a and b are consecutive integers

q: sum of a+b must be an odd number

Assume that p is true

That is a and b are consecutive integers.

$$b = a+1$$

$$\text{Then } a+b = a+a+1 = 2a+1$$

We know that any number multiplied by a 2 must be even. We also know that if we add 1 to any even number, it becomes odd.

So a+b is odd.

That gives q is true.

Hence $p \rightarrow q$ is true.

The given statement is thus proved by a direct proof.

Indirect Proof

For the a conditional $p \rightarrow q$ its contrapositive is $\neg q \rightarrow \neg p$.

They are logically equivalent.

The indirect method of proving a conditional $p \rightarrow q$ has the following lines of argument

1) Assume that $\neg q$ is true

2) Starting with the hypothesis and employing the rules/laws of logic and other known facts, infer that $\neg p$ is true.

3) $\neg q \rightarrow \neg p$ is true. This is equivalent to $p \rightarrow q$ is true.

Example 1:

Give a indirect proof of the statement Let n be an integer.

"if n^2 is odd, then n is odd "

solution:

Here, the conditional to be proved is:

"if n^2 is odd, then n is odd "

Here,

p : n^2 is odd.

q : n is odd.

$\neg p$: n^2 is not odd.

$\neg q$: n is not odd.

We first prove that the contrapositive is true.

Assume that $\neg q$ is true

That is, n is not an odd integer.

Then $n = 2k$, where k is an integer.

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2)$$

We know that any number multiplied by an 2 must be even. So that n^2 is even.

That means n^2 is not odd.

That is, $\neg p$ is true.

This proves the contrapositive statement $\neg q \rightarrow \neg p$ is true.

This is equivalent to $p \rightarrow q$ is true.

Example 2:

Give an indirect proof of the statement:

"The product of two even integers is an even integer".

Solution:

"The product of two even integers is an even integer"

The given statement is equivalent to the following statement:
"If a and b are even integers, then ab is an even integer".

Here,

p : a and b are even integers,

and q : ab is an even integer.

$\neg p$: a and b are not even integers

$\neg q$: ab is not even integer

We first prove that the contrapositive $\neg q \rightarrow \neg p$ is true.

Assume $\neg q$ is true.

That is ab is not even integer or ab is odd integer.

This means that ab is not divisible by 2.

Hence a is not divisible by 2, and b is not divisible by 2.

That is, a is not an even integer and b is not an even integer.

This means that a and b are not even integers is true.

This gives $\neg p$ is true.

This proves the contrapositive statement $\neg q \rightarrow \neg p$ is true.

This is equivalent to $p \rightarrow q$ is true.

Example 3:

For each of the following statements, provide an indirect proof by stating proving the contrapositive of the given statement.

(1) For all integers k and l , if kl is odd, then both k and l are odd

(2) For all integers k and l , if $k+l$ is even, then k and l are both even or both odd.

Solution:

(1) For all integers k and l , if kl is odd, then both k and l are odd.

We now prove this by contrapositive.

Here p : kl is odd

q : both k and l are odd

$\neg p$: kl is not odd

$\neg q$: both k and l are not odd

Assume that $\neg q$ is true. That is both k and l are not odd.

If k is even, then $k = 2m$ for some integer m ,

and $kl = (2m)l = 2(ml)$

Similarly if l is even, then $kl = k(2n) = 2(kn)$ for some integer

n . So that kl is even, means kl is not odd.

this gives $\neg p$ is true.

This proves the contrapositive $\neg q \rightarrow \neg p$ is true.

This is equivalent to $p \rightarrow q$ is true.

(2) For all integers k and l , if $k+l$ is even, then k and l are both even or both odd.

We now prove this by contrapositive.

Here p : $k+l$ is even

q : both k and l are even or odd

$\neg p$: $k+l$ is not even

$\neg q$: k and l are not both even or not both odd

Assume that $\neg q$ is true. That is k and l are not both even or not both odd, means one of k and l is odd and other is even.

Suppose k is odd and l is even.

Then,

$$k=2m+1$$

$$l=2n$$

Consequently, $k+l=2m+1+2n=2(m+n)+1$

.If k is even and l is odd.

Then, $k=2m$

$$l=2n+1$$

Consequently, $k+l=2m+2n+1=2(m+n)+1$.

We know that any number multiplied by a 2 must be even. We also know that if we add 1 to any even number, it becomes odd.

So $k+l$ is odd or $k+l$ is not

even. this gives $\neg p$ is true.

This proves the contrapositive $\neg q \rightarrow \neg p$ is true.

This is equivalent to $p \rightarrow q$ is true.

Proof by Contradiction

The lines of argument in this method of proof of the statement $p \rightarrow q$ are as follows:

1. Hypothesis: Assume that $p \rightarrow q$ is false. That is, assume that p is true and q is false.
2. Analysis: Starting with the hypothesis that q is false and employing the rules of logic and other known facts, infer that p is false. This contradicts the assumption that p is true.
3. Conclusion: Because of the contradiction arrived in the analysis, we infer that $p \rightarrow q$ is true.

Example 1:

Provide a proof by contradiction of the following statement:

For every integer n , if n^2 is odd, then n is odd.

Solution:

Let n be any integer. Then the given statement reads $p \rightarrow q$, where

p : n^2 is odd, and

q : n is odd.

Assume that $p \rightarrow q$ is false; that is, assume that p is true and q is false.

Now, q is false means: n is even, so that $n = 2k$ for some integer k .

Then $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$

We know that any number multiplied by an 2 must be even.

That is n^2 even, means p is false.

This contradicts the assumption that p is true.

In view of this contradiction, we infer that the given conditional $p \rightarrow q$ is true .

Example 2:

Prove the statement

"The square of an even integer is an even integer" by the method of contradiction

Solution:

The given statement is equivalent to the following statement

"if n is an even integer, then n^2 is an even integer".

p : n is an even integer,

and q : n^2 is an even integer.

Assume that $p \rightarrow q$ is false, that is assume that p is true and q is false.

Since q is false, that is, n^2 is not an even integer.

Therefore, $n^2 = nxn$ is not divisible by 2.

This implies that n is not divisible by 2. That is, n is not an even integer.

This means that p is false, which contradicts the assumption that p is true.

In view of this contradiction, we infer that the given proposition $p \rightarrow q$ is true.

Example 3:

Prove that if m is an even integer, then $m+7$ is an odd integer

Solution:

Here, the given statement is $p \rightarrow q$,

where p : m is even.

q : $m+7$ is odd

Assume that $p \rightarrow q$ is false; that is, assume that p is true and q is false.

Since q is false, $m+7$ is even. Hence, $m+7 = 2k$ for some integer k .

This yields

$$m = 2k - 7 = (2k - 8) + 1 = 2(k - 4) + 1$$

which shows that m is odd.

This means that p is false, which contradicts the assumption that p is true. In

view of this contradiction, we infer that the given statement $p \rightarrow q$ is true.

Example 1 :

Give (1) a direct proof, (2) an indirect proof, and (3) proof by contradiction, for the following statement:

"If n is an odd integer, then $n+9$ is an even integer"

Solution:

(1) Direct proof:

Assume that n is an odd integer.

Then $n = 2k + 1$ for some integer k .

This gives $n+9 = (2k+1)+9 = 2k+10 = 2(k+5)$.

We know that any number multiplied by a 2 must be even.

So $n+9$ is even.

This establishes the truth of the given statement by a direct proof.

(2) Indirect proof:

Assume that $n+9$ is not an even integer.

Then $n+9 = 2k + 1$ for some integer k .

This gives $n = (2k+1)-9 = 2k-8 = 2(k-4)$,

which shows that n is even.

This means that n is not odd.

Thus, if $n+9$ is not even integer, then n is not odd. This proves the contrapositive of the given statement.

This proof of the contrapositive serves as an indirect proof of the given statement.

(3) Proof by contradiction:

Assume that the given statement is false. That

is, assume that n is odd and $n+9$ is odd.

Since $n+9$ is odd, $n+9 = 2k + 1$ for some integer k

so that $n = (2k+1)-9 = 2k-8 = 2(k-4)$ which shows that n is even. This contradicts the assumption that n is odd.

Hence the given statement must be true.

Example 2:

Give a proof by contradiction of the theorem “If $3n + 2$ is odd, then n is odd.”

Solution:

Let p : “ $3n + 2$ is odd”

and q be “ n is odd.”

To construct a proof by contradiction, assume that both p is true and q is false.

That is, assume that $3n + 2$ is odd and that n is not odd.

Because n is not odd, we know that it is even.

Because n is even, there is an integer k such that $n = 2k$. This implies that $3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1)$. We know that any number multiplied by an 2 must be even.

So $3n + 2$ is even, it means $3n + 2$ is not odd.

This implies p is false.

This contradicts the assumption that p is true .

Hence the given statement must be true.

Proof by Exhaustion or Proof by cases:

If S consists of only a limited number of elements, we can prove that the statement " $\forall x \in S P(x)$ " is true by considering $p(a)$ for each a in S and verifying that $p(a)$ is true (in each case). Such a method of proof is called the method of exhaustion".

Example 1:

Prove that every even integer n with $2 \leq n \leq 26$ can be written as a sum of at most three perfect squares.

Solution:

Let $S = \{2, 4, 6, \dots, 24, 26\}$.

We have to prove that the statement is true, where $p(x)$: x is a sum of at most three perfect squares. We observe the following:

$2 = 1^2 + 1^2$	$4 = 2^2$
$6 = 2^2 + 1^2 + 1^2$	$8 = 2^2 + 2^2$
$10 = 3^2 + 1^2$	$12 = 2^2 + 2^2 + 2^2$
$14 = 3^2 + 2^2 + 1^2$	$16 = 4^2$
$18 = 4^2 + 1^2 + 1^2$	$20 = 4^2 + 2^2$
$22 = 3^2 + 3^2 + 2^2$	$24 = 4^2 + 2^2 + 2^2$
$26 = 5^2 + 1^2$	

The above facts verify that each x in S is a sum of at most three perfect squares. This establishes the required result.

Example 2:

Prove that $(n + 1)^3 \geq 3^n$ if n is a positive integer with $n \leq 4$.

Solution:

We use a proof by exhaustion.

We only need verify the inequality $(n + 1)^3 \geq 3^n$ when $n = 1, 2, 3$, and 4 .

For $n = 1$, we have $(n + 1)^3 = 2^3 = 8$ and $3^n = 3^1 = 3$;

for $n = 2$, we have $(n + 1)^3 = 3^3 = 27$ and $3^n = 3^2 = 9$;

for $n = 3$, we have $(n + 1)^3 = 4^3 = 64$ and $3^n = 3^3 = 27$;

and for $n = 4$, we have $(n + 1)^3 = 5^3 = 125$ and $3^n = 3^4 = 81$. In each of these four cases, we see that $(n + 1)^3 \geq 3^n$.

We have used the method of exhaustion to prove that $(n + 1)^3 \geq 3^n$ if n is a positive integer with $n \leq 4$.

Example 3:

Prove that if n is an integer, then $n^2 \geq n$.

Solution:

We can prove that $n^2 \geq n$ for every integer by considering three cases, when $n =$

0, when $n \geq 1$, and when $n \leq -1$.

We split the proof into three cases because it is straightforward to prove the result by considering zero, positive integers, and negative integers separately.

Case (i): When $n = 0$, because $0^2 = 0$, we see that $0^2 \geq 0$. It follows that $n^2 \geq n$ is true in this case.

Case (ii): When $n \geq 1$,

when we multiply both sides of the inequality $n \geq 1$ by the positive integer n , we obtain $n \cdot n \geq n \cdot 1$.

This implies that $n^2 \geq n$ for $n \geq 1$.

Case (iii): In this case $n \leq -1$.

However, $n^2 \geq 0$.

It follows that $n^2 \geq n$.

Because the inequality $n^2 \geq n$ holds in all three cases, we can conclude that if n is an integer, then $n^2 \geq n$.

Proof of Existence

If S is domain, we can prove that the statement " $\exists x \in S P(x)$ " is true by considering if any one element $a \in S$ such that $p(a)$ is true. Such a method of proof is called the method of existence.

Example 1 :

Prove that there exists a real number x such that $x+1=0$.

Solution:

It is sufficient to exhibit one real number x such that $x+1=0$. We check the $x = -1$ is one such real number.

Example 2 :

Prove that there exists a real number x such that $x^3+2x^2-5x-6=0$

Solution:

It is sufficient to exhibit one real number x such that $x^3+2x^2-5x-6=0$. We check the $x = -1$ is one such real number.

Example 3:

Prove that there exist positive integers m and n such that m , n and $m+n$ are all perfect squares.

solution:

We note that $m = 9$ and $n = 16$ are perfect squares and so is $m+n=9+16=25$.

This proves the required result.

BASIC STRUCTURES

Sets:

Sets are collection of well-defined distinct objects.

A set is represented by a capital letters.

Elements of set are usually represented by small letters.

Examples:

- 1) Collection of numbers
- 2) Collection of days of a week
- 3) Collection of vowels in English alphabets
- 4) Collection of prime numbers less than 5

Some standard sets are:

- Set of natural numbers, $\mathbb{N} = \{1, 2, 3, \dots\}$
- Set of whole numbers, $\mathbb{W} = \{0, 1, 2, 3, \dots\}$
- Set of integers, $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- Set of rational numbers, $\mathbb{Q} = \{p/q \mid q \text{ is an integer and } q \neq 0\}$
- Set of irrational numbers, $\mathbb{Q}' = \{x \mid x \text{ is not rational}\}$
- Set of real numbers, $\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}'$

Example: The collection of even natural numbers less than 10 can be represented in the form of a set, $A = \{2, 4, 6, 8\}$.

Members or elements of sets:

The objects which make up a set are called its members or elements.

The elements of a set are enclosed in curly brackets separated by commas.

To denote that an element is contained in a set, the symbol ' $\in$ ' is used.

In the above example, $2 \in A$. If an element is not a member of a set, then it is denoted using the symbol ' $\notin$ '. For example, $3 \notin A$.

Cardinality of sets:

The cardinal number, cardinality, or order of a set denotes the total number of elements in the set.

For natural even numbers less than 10,

$$A = \{2, 4, 6, 8\}$$

$$n(A) = 4$$

Representation of set:

1) Roster Form or Tabular form:

Roster notation in which the elements of the sets are enclosed in curly brackets separated by commas.

For example,

Set of the first five even numbers in a roster form

$$\text{Set } B = \{2, 4, 6, 8, 10\}$$

Finite Roster Notation of Sets:

Set of first five natural numbers

$$A = \{1, 2, 3, 4, 5\}$$

Infinite Roster Notation of Sets:

Set of multiples of 5

$$\text{Set } B = \{5, 10, 15, 20, \dots\}$$

Set-Builder form or rule method:

If the elements of a set have a common property then they can be defined by describing the property.

For example,

The elements of the set $A = \{1, 2, 3, 4, 5, 6\}$ have a common property, which states that all the elements in the set A are natural numbers less than 7. No other natural numbers retain this property. Hence, we can write the set A as follows:

$A = \{x \mid x \text{ is a natural number less than } 7\}$ which can be read as “ A is the set of elements x such that x is natural numbers less than 7”.

The above set can also be written as $A = \{x \mid x \in \mathbb{N}, x < 7\}$.

We can also write, set $A = \{x \mid x \text{ is natural numbers less than } 7\}$.

Types of sets:**Empty Sets:**

The set, which has no elements, is also called a null set or void set. It is denoted by $\{\}$ or $\emptyset$.

Below are the two examples of the empty set.

1) $A = \{x \mid 9 < x < 10, x \text{ is a natural number}\}$ will be a null set because there is no natural number between numbers 9 and 10.

2) $B = \{x \mid x \text{ is whole number less than } 0\}$

Singleton Sets:

The set which has just one element is named a singleton set.

For example,

Set 1) $A = \{8\}$ is a singleton set.

2) $B = \{x \mid x \text{ is natural number less than } 2\}$

3) $C = \{x \mid x \text{ is even prime number}\}$

Finite and Infinite Sets:

Any set which is empty or contains a definite and countable number of elements is called a **finite set**.

Sets defined otherwise, for uncountable or indefinite numbers of elements are referred to as **infinite sets**.

For example,

$A = \{3, 4, 5, 6, 7\}$ is a finite set, as it has a finite number of elements.

$B = \{a, e, i, o, u\}$ is a finite set because it represents the vowel letters in the English alphabetical series.

$C = \{p \mid p \text{ is a prime number}\}$ is an infinite set.

$D = \{k \mid k \text{ is a real number}\}$ is also an infinite set.

Equal Sets:

If every element of set A is also the elements of set B and if every element of set B is also the elements of set A , then sets A and B are called equal sets.

It means set A and set B have equivalent elements and that we can denote it as: $A = B$

For example, let $A = \{3, 4, 5, 6\}$ and $B = \{6, 5, 4, 3\}$, then $A = B$

$A = \{\text{set of even numbers}\}$ and $B = \{\text{set of natural numbers}\}$ then $A \neq B$, because natural numbers consist of all the positive integers starting from 1, 2, 3, 4, 5 to infinity, but even numbers start with 2, 4, 6, 8, and so on.

Equivalent sets:

If A and B are two sets and $A \not\subseteq B$ and $B \not\subseteq A$ but both the sets have equal number of elements then the set A and B are called equivalent.

Subsets:

A set B is said to be a subset of set A if the elements of set B belong to set A , or you can say each element of set B is present in set A .

Subset of a set is denoted by the symbol ($\subseteq$) and written as $B \subseteq A$.

Each set is a subset of its own set, and a void set or empty set is a subset of all sets.

$A = \{1, 2, 3\}$

Then $\{1, 2\} \subseteq A$.

Similarly, other subsets of set A are: $\{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}, \{1,2,3\}, \{\}$.

Proper Subset:

If $A \subseteq B$ and $A \neq B$, then A is called the proper subset of B and it can be written as $A \subset B$.

Or A set is proper subset of a set B, if

i) $A \subseteq B$

ii) B contains at least one element that is not in A.

Example 1: If $A = \{1, 2, 3\}$ and $B = \{1, 2\}$

i) $B \subseteq A$

ii) B contains at least one element that is not in A.

So B is proper subset of A

Example 2: If $A = \{2, 5, 7\}$ is a subset of $B = \{2, 5, 7\}$ then it is not a proper subset of $B = \{2, 5, 7\}$

But, $A = \{2, 5\}$ is a subset of $B = \{2, 5, 7\}$ and is a proper subset also.

Superset:

Set A is said to be the superset of B if all the elements of set B are the elements of set A. It is represented as $A \supset B$.

For example, if set $A = \{1, 2, 3, 4\}$ and set $B = \{1, 3, 4\}$, then set A is the superset of B.

Power Sets:

The set of all subsets is known as power sets. We know the empty set is a subset of all sets, and each set is a subset of itself. Taking an example of set $X = \{2, 3\}$. From the above-given statements, we can write,

$\{\}$ is a subset of $\{2, 3\}$

$\{2\}$ is a subset of $\{2, 3\}$

$\{3\}$ is a subset of $\{2, 3\}$

$\{2, 3\}$ is also a subset of $\{2, 3\}$

Therefore, power set of $X = \{2, 3\}$ is $P(X) = \{\{\}, \{2\}, \{3\}, \{2, 3\}\}$

Universal Sets:

A set that contains all the elements of other sets is called a universal set.

Generally, it is represented as 'U.'

For example, set $A = \{1, 2, 3\}$, set $B = \{3, 4, 5, 6\}$, and $C = \{5, 6, 7, 8, 9\}$.

Then, we will write the universal set as, $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

Disjoint Sets:

If two sets X and Y do not have any common elements, and their intersection results in zero(0), then set X and Y are called disjoint sets. It can be represented as; $X \cap Y = 0$.

Venn diagrams:

Relations between sets can be depicted in diagrams called Venn diagrams.

For example:

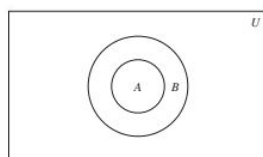


FIGURE 2 Venn diagram showing that A is a subset of B.

Operations on sets:

Union of Sets

The union of two sets consists of all their elements. It is denoted by $(\cup)$.

Or Set consisting of all the elements that belong to A or B is called union of A and B.

Is denoted by $A \cup B$

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

For example, set $A = \{2, 3, 7\}$ and set $B = \{4, 5, 8\}$.

Then the union of set A and set B will be:

$$A \cup B = \{2, 3, 7, 4, 5, 8\}$$

Venn diagram for union:

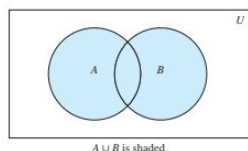


FIGURE 1 Venn diagram of the union of A and B.

Intersection of Sets

The set of all elements, which are common to all the given sets, gives an intersection of sets. It is denoted by $\cap$.

Or Set consisting of all the elements that belong to A and B is called intersection of A and B.

Is denoted by $A \cap B$

For example, set $A = \{2, 3, 7\}$ and set $B = \{2, 4, 9\}$.

So, $A \cap B = \{2\}$

Venn diagram for intersection:

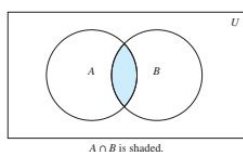


FIGURE 2 Venn diagram of the intersection of A and B.

Complement of a Set:

A universal set U and set $A \subseteq U$ then

Set consisting of all the elements that belong to U but not in A is called Complement of a set A (in U).

Is denoted by A or A' or A^c

$$A^c = \{x \mid x \in U \text{ but } x \notin A\}.$$

Venn diagram for complement:

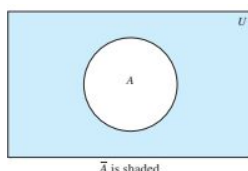


FIGURE 4 Venn diagram for the complement of the set A.

Example: If $A = \{1, 2, 3, 4\}$ and $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ then find A complement (A).

Solution:

$A = \{1, 2, 3, 4\}$ and Universal set $= U = \{1, 2, 3, 4, 5, 6, 7, 8\}$

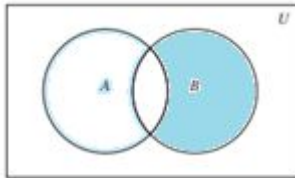
Complement of set A contains the elements present in universal set but not in set A. Elements are 5, 6, 7, 8.

$\therefore A \text{ complement} = A = \{5, 6, 7, 8\}$

Relative Complement:

Given two sets A and B, the set of all elements that belong to it but not to A is called the Complement of A relative to B (or relative complement of A in B) and is denoted by $B-A$ that is

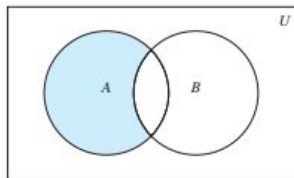
$B-A = \{x / x \in B \text{ and } x \notin A\}$



B-A is shaded

The set $A-B$ is defined similarly the set of all elements that belong to it but not to B is called the complement of B relative to A (or relative complement of B in A) and is denoted by $A-B$ that is

$A-B = \{x / x \in A \text{ and } x \notin B\}$.



A-B is shaded.

FIGURE 3 Venn diagram for the difference of A and B.

Note that $B-A$ is not the same as $A-B$.

Example: If set $A = \{a, b, c, d\}$ and $B = \{b, c, e, f\}$ then, find $A-B$.

Solution: Let's find the difference of the two sets,

$A - B = \{a, d\}$ and $B - A = \{e, f\}$.

Example: if $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 7, 8\}$, then find $B-A$ and $A-B$

Solution:

$B-A = \{7, 8\}$ and $A-B = \{1, 2\}$.

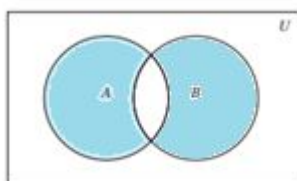
Symmetric Difference:

For two sets A and B, the relative complement of $A \cap B$ in $A \cup B$ is called the symmetric difference of A and B and is denoted by $A \Delta B$. Thus,

$A \Delta B = (A \cup B) - (A \cap B) = \{x | x \in A \cup B \text{ and } x \notin A \cap B\}$

Or

$A \Delta B = (A - B) \cup (B - A) = \{x | x \in A - B \text{ or } x \in B - A\}$



$A \Delta B$ is shaded

Example:

If $A = \{a, b, c, k, m, n\}$ and $B = \{c, n\}$

$A \cup B = \{a, b, c, k, m, n\}$ and $A \cap B = \{c, n\}$

So, the symmetric difference between the given sets A and B is

$A \Delta B = \{a, b, k, m\}$.

FUNCTIONS

Cartesian product:

Let A and B be sets. The Cartesian product of A and B , denoted by $A \times B$, is the set of all ordered pairs (a, b) , where $a \in A$ and $b \in B$. Hence,

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

Example: What is the Cartesian product of $A = \{1, 2\}$ and $B = \{a, b, c\}$?

Solution: The Cartesian product $A \times B$ is

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}.$$

Functions:

Let A and B be two nonempty sets. Then a function f from A to B is a relation from A to B such that each a in A there is a unique b in B such that $(a, b) \in f$.

Or

Let A and B be nonempty sets. A function f from A to B is an assignment of each element of A to exactly one element of B .

We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A . If f is a function from A to B , we write $f: A \rightarrow B$.

Note: Functions are sometimes also called **mappings** or **transformations**.

Also, if f is a function from A to B , we say that f maps A to B .

If f is a function from A to B , we say that A is the **domain** of f and B is the **codomain** of f . If $f(a) = b$, we say that **b** is the **image** of a and **a** is a **preimage** of b .

Range:

The range of f is the set of all images of elements of A .

Or The subset of B consisting of the images of all elements of A under f is called range.

Note: Let f_1 and f_2 be functions from A to $\mathbf{R}$. Then $f_1 + f_2$ and $f_1 f_2$ are also functions from A to $\mathbf{R}$ defined for all $x \in A$ by

$$(f_1 + f_2)(x) = f_1(x) + f_2(x),$$

$$(f_1 f_2)(x) = f_1(x)f_2(x).$$

Example: Let f_1 and f_2 be functions from $\mathbf{R}$ to $\mathbf{R}$ such that $f_1(x) = x^2$ and $f_2(x) = x - x^2$.

What are the functions $f_1 + f_2$ and $f_1 f_2$?

Solution: From the definition of the sum and product of functions, it follows that $(f_1 + f_2)(x) = f_1(x) + f_2(x) = x^2 + (x - x^2) = x$ and

$$(f_1 f_2)(x) = x^2(x - x^2) = x^3 - x^4.$$

Types of function:

Identity function:

A function $f: A \rightarrow A$ is an identity function if the image of every element of A is itself.

Or

A function $f: A \rightarrow A$ such that $f(a)=a$ for every $a \in A$ is called identity function on A

Constant function:

A function $f: A \rightarrow B$ is a constant function if all elements of A have same image in B .

Or

A function $f: A \rightarrow B$ such that $f(a)=c$ for every $a \in A$, where c is a fixed element of B , is called a constant function.

One-to-one, or an injective function:

A function f is said to be one-to-one, or an injective function, if and only if $f(a) = f(b)$ implies that $a = b$ or $f(a) \neq f(b)$ implies that $a \neq b$ for all a and b in the domain of f .

Example: Determine whether the function f from $\{a, b, c, d\}$ to $\{1, 2, 3, 4, 5\}$ with $f(a) = 4$, $f(b) = 5$, $f(c) = 1$, and $f(d) = 3$ is one-to-one.

Solution: The function f is one-to-one because f takes on different values at the four elements of its domain.

Onto function or surjective function:

A function f from A to B is called onto, or a surjection, if and only if every element in $b \in B$, there exist at least element $a \in A$ such that $f(a) = b$.

Example: Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3\}$ defined by $f(a) = 3$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$. Is f an onto function?

Solution: Because all three elements of the codomain are images of elements in the domain, we see that f is onto. This is illustrated in Figure 4. Note that if the codomain were $\{1, 2, 3, 4\}$, then f would not be onto.

One to One correspondence:

The function f is a one-to-one correspondence, or a bijection, if it is both one-to-one and onto. We also say that such a function is **bijective**.

Example: Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3, 4\}$ with $f(a) = 4$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$. Is f a bijection?

Solution: The function f is one-to-one and onto. It is one-to-one because no two values in the domain are assigned the same function value. It is onto because all four elements of the codomain are images of elements in the domain. Hence, f is a bijection.

Into function: For a function $f: A \rightarrow B$ to be an into function there will be one or more elements in set B that do not have a pre-image in set A .

Many to one function: A function f is said to be many to-one function, such that $a \neq b$ implies that $f(a) = f(b)$ for all $a, b \in A$ and $f(a), f(b) \in B$.

Floor and Ceiling functions

Let x be any real number. Then x is an integer or x lies between two integers. Let $\lfloor x \rfloor$ denote the greatest integer that is less than or equal to x , and $\lceil x \rceil$ denote the least integer that greater than or equal to x . Then $\lfloor x \rfloor$ is called the floor of x and $\lceil x \rceil$ is called the ceiling of x . For example (floor), $\lfloor 2.4 \rfloor = 2$, $\lfloor -2.4 \rfloor = -3$, and for ceiling; $\lceil 2.4 \rceil = 3$, and $\lceil -2.4 \rceil = -2$.

The function from the set of all real numbers to the set of all integers under which a real number x has $\lfloor x \rfloor$ as its image is called the **floor function (or the greatest integer function)**. Similarly, the function from the set of all real numbers to the set of all integers under which a real number x has $\lceil x \rceil$ as its image is called the **ceiling function (or the least integer function)**. If f denotes the floor function, we have, for example,

$$f(2) = \lfloor 2 \rfloor = 2$$

$$f(2.1) = \lfloor 2.1 \rfloor = 2$$

$$f(-6.3) = \lfloor -6.3 \rfloor = -7$$

If c denotes the ceiling function, we have, for example,

$$c(2) = \lceil 2 \rceil = 2$$

$$c(2.1) = \lceil 2.1 \rceil = 3$$

$$c(-6.3) = \lceil -6.3 \rceil = -6.$$

Composition of function:

Let f be a function from the set A to the set B and let g be a function from the set B to the set C . The composition of the functions f and g , denoted for all $a \in A$ by $(g \circ f)$, is the function from A to C defined by

$$(g \circ f)(a) = g(f(a)).$$

Example: Let g be the function from the set $\{a, b, c\}$ to itself such that $g(a) = b$, $g(b) = c$, and $g(c) = a$. Let f be the function from the set $\{a, b, c\}$ to the set $\{1, 2, 3\}$ such that $f(a) = 3$, $f(b) = 2$, and $f(c) = 1$. What is the composition of f and g , and what is the composition of g and f ? **Solution:** The composition $f \circ g$ is defined by

$$(f \circ g)(a) = f(g(a)) = f(b) = 2,$$

$$(f \circ g)(b) = f(g(b)) = f(c) = 1,$$

$$\text{and } (f \circ g)(c) = f(g(c)) = f(a) = 3.$$

Note that $g \circ f$ is not defined, because the range of f is not a subset of the domain of g .

Example: Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g ? What is the composition of g and f ?

Solution: Both the compositions $f \circ g$ and $g \circ f$ are defined. Moreover,

$$(f \circ g)(x) = f(g(x)) = f(3x + 2) = 2(3x + 2) + 3 = 6x + 7$$

and

$$(g \circ f)(x) = g(f(x)) = g(2x + 3) = 3(2x + 3) + 2 = 6x + 11.$$

Invertible function:

Let f be a one-to-one correspondence from the set A to the set B . The inverse function of f is the function that assigns to an element b belonging to B the unique element a in A such that $f(a) = b$. The inverse function of f is denoted by f^{-1} . Hence, $f^{-1}(b) = a$ when $f(a) = b$.

Example: Let f be the function from $\{a, b, c\}$ to $\{1, 2, 3\}$ such that $f(a) = 2$, $f(b) = 3$, and $f(c) = 1$. Is f invertible, and if it is, what is its inverse?

Solution: The function f is invertible because it is a one-to-one correspondence.

The inverse function f^{-1} reverses the correspondence given by f , so $f^{-1}(1) = c$, $f^{-1}(2) = a$, and $f^{-1}(3) = b$.

MATRICES

A set of m, n numbers (real or complex), arranged in a rectangular formation (array or table) having m rows and n columns and enclosed by a square bracket [] is called m x n matrix (read "m by n matrix").

An m x n matrix is expressed as

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ . & . & . & . \\ . & . & . & . \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

the matrix A is sometimes denoted by simplified form as

The matrix A is sometimes denoted by simplified form as $A = [a_{ij}]$

Matrices are usually denoted by capital letters A, B, C etc.. and its elements by small letters a,b,c etc.

Note: 1) The horizontal arrays (- ->) of a matrix are called its rows.

2) The vertical arrays (↓) are called its columns.

Order of matrix: A matrix having m rows and n columns is said to have the **order of matrix** and it is denoted m×n (read as m by n)

Example: A, B and C matrices of order (2× 3), (3 ×2) and (3× 3) respectively.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 7 & 9 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 6 \\ 6 & 10 \\ 5 & 14 \end{bmatrix}$$

$$C = \begin{bmatrix} 5 & 8 & 9 \\ 6 & 5 & 14 \\ 14 & 4 & 15 \end{bmatrix}$$

Types of matrices:

Zero Matrix or Null Matrix: Any matrix in which all the components are zero is termed a zero matrix. It is also recognized as the null matrix and is denoted by O.

Row Matrix : Any matrix which contains one row and n columns is said to be a row matrix.

Column Matrix: Any matrix which contains m rows and one column is stated to be a column matrix.

Square Matrix: Any matrix in which the number of rows is equivalent to the number of columns.

Rectangular Matrix: A matrix is identified as a rectangular matrix if the number of rows is not identical to the number of columns.

Diagonal Matrix: Any square matrix at least one element of the principal diagonal is non- zero and all the other elements are zero is called a diagonal matrix.

Scalar Matrix: A diagonal matrix in which all the principal diagonal elements are the same.

Identity Matrix or Unit Matrix: A diagonal matrix in which all the principal diagonal components are identical to 1 is called an identity matrix or unit matrix. An identity matrix of order n is denoted by I.

Triangular matrix: A square matrix is said to be triangular matrix, if all the elements above or below the principle diagonal are zero.

If all the elements above the principle diagonal are zero the matrix is lower triangular matrix.

If all the elements below the principle diagonal are zero matrix is upper triangular matrix.

Sub matrix: The matrix formed by deleting of some rows or columns from a matrix is called sub matrix.

Equal matrix:

Two matrices are equal if they have the same order and the corresponding elements are identical.

Equivalent matrix: Two matrices are having same order then it is called equivalent matrices.

Transpose of a Matrix: The transpose of a matrix can be achieved by interchanging its rows into columns and its columns into rows.

Symmetric Matrix: A symmetric matrix is identified as the square matrix that is equivalent to its transpose matrix.

Or in other words if A is a square matrix such that ${}^T(\text{transpose of } A) = A$ then A is said to be a symmetric matrix.

Skew symmetric Matrix: Any square matrix $A = [a_{ij}]$ is said to be a skew-symmetric matrix if and only if $a_{ij} = -a_{ji}$, $\forall i$ and j and principle diagonal elements are zero.

Or in other words we can say that if A is a square matrix such that ${}^T(\text{transpose of } A) = (-A)$ then A is said to be a skew-symmetric matrix.

Zero one matrix: A zero-one matrix has entries that are either zero or one.

The arithmetic is based on the Boolean operations $\wedge$ and $\vee$, which operate on pairs of bits, defined by

$$b_1 \wedge b_2 = \begin{cases} 1 & \text{if } b_1 = b_2 = 1 \\ 0 & \text{otherwise,} \end{cases}$$

$$b_1 \vee b_2 = \begin{cases} 1 & \text{if } b_1 = 1 \text{ or } b_2 = 1 \\ 0 & \text{otherwise} \end{cases}$$

Join of two matrix:

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be $m \times n$ zero-one matrices. Then the join of A and B is the zero-one matrix with (i, j) th entry $a_{ij} \vee b_{ij}$. The join of A and B is denoted by $A \vee B$.

Meet of two matrix:

The meet of A and B is the zero-one matrix with (i, j) th entry $a_{ij} \wedge b_{ij}$. The meet of A and B is denoted by $A \wedge B$.

Boolean product:

Let $A = [a_{ij}]$ be an $m \times k$ zero-one matrix and $B = [b_{ij}]$ be a $k \times n$ zero-one matrix. Then the Boolean product of A and B, denoted by $A \odot B$, is the $m \times n$ matrix with (i, j) th entry c_{ij} where

$$c_{ij} = (a_{i1} \wedge b_{1j}) \vee (a_{i2} \wedge b_{2j}) \vee \cdots \vee (a_{ik} \wedge b_{kj})$$

Addition of matrices:

Let A and B be two matrices of same order, then their sum $A+B$ is a matrix whose elements are sum of the corresponding elements of A and B.

Subtraction of matrices (difference of matrices):

Let A and B be two matrices of same order, then their difference $A-B$ is a matrix whose elements are difference of the corresponding elements of A and B.

Scalar multiplication:

Let A is matrix and K is any scalar then product KA is a scalar multiplication obtained by multiplying each element of A by K .

Multiplication of Matrices:

The product of two matrices is defined only when number of columns of the first matrix is equal to number of rows in the second matrix.

If A is matrix of order $m \times p$ and B is matrix of order $p \times n$ then matrix multiplication is possible as number of columns in first matrix and number of rows in second matrix are equal then the product AB of order $m \times n$ is the sum of the product of elements of row of A with corresponding elements of column B .

SEQUENCES AND SUMMATIONS

Sequences:

A sequence is a function from a subset of the set of integers (usually either the set $\{0, 1, 2, \dots\}$ or the set $\{1, 2, 3, \dots\}$) to a set S .

We use the notation a_n to denote the image of the integer n .

We call a_n is a term of the sequence.

Example: Consider the sequence $\{a_n\}$, where –

$$a_n = \frac{1}{n}$$

The list of the terms of this sequence, beginning with a_1 , namely,

$a_1, a_2, a_3, a_4, \dots$ starts with

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$$

Arithmetic progression:

An arithmetic progression is a sequence of the form

$a, a + d, a + 2d, \dots, a + nd, \dots$

where the initial term a and the common difference d are real numbers.

Example: The sequences $\{s_n\}$ with $s_n = -1 + 4n$ and if we start at $n = 0$. The list of terms $s_0, s_1, s_2, s_3, \dots$ begins with $-1, 3, 7, 11, \dots$

progressions with initial terms and common differences equal to -1 and 4 ,

The sequences $\{t_n\}$ with $t_n = 7 - 3n$ are both arithmetic

if we start at $n = 0$ and the list of terms $t_0, t_1, t_2, t_3, \dots$ begins with

$7, 4, 1, -2, \dots$

progressions with initial terms and common differences equal to 7 and -3 , respectively.

Geometric progression:

A geometric progression is a sequence of the form

$a, ar, ar^2, \dots, ar^n, \dots$

where the initial term a and the common ratio r are real numbers.

Example: The sequences $\{b_n\}$ with $b_n = (-1)^n$,

The list of terms $b_0, b_1, b_2, b_3, b_4, \dots$ begins with

$1, -1, 1, -1, 1, \dots$

geometric progressions with initial term and common ratio equal to 1 and -1 $\{c_n\}$

with $c_n = 2 \cdot 5^n$,

the list of terms $c_0, c_1, c_2, c_3, c_4, \dots$ begins with

$2, 10, 50, 250, 1250, \dots$

Recurrence relation

A recurrence relation for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence, namely, $a_0, a_1, \dots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integer. A sequence is called a solution of a recurrence relation if its terms satisfy the recurrence relation.

Example: Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 1, 2, 3, \dots$, and suppose that $a_0 = 2$. What are a_1 , a_2 , and a_3 ?

Solution: We see from the recurrence relation that $a_1 = a_0 + 3 = 2 + 3 = 5$. It then follows that $a_2 = 5 + 3 = 8$ and $a_3 = 8 + 3 = 11$.

Example: Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} - a_{n-2}$ for $n = 2, 3, 4, \dots$, and suppose that $a_0 = 3$ and $a_1 = 5$. What are a_2 and a_3 ?

Solution: We see from the recurrence relation that $a_2 = a_1 - a_0 = 5 - 3 = 2$ and $a_3 = a_2 - a_1 = 2 - 5 = -3$. We can find a_4 , a_5 , and each successive term in a similar way.

Fibonacci sequence:

The Fibonacci sequence, $f_0, f_1, f_2, \dots$, is defined by the initial conditions

$f_0 = 0$, $f_1 = 1$, and the recurrence relation

$f_n = f_{n-1} + f_{n-2}$ for $n = 2, 3, 4, \dots$.

Example: Find the Fibonacci numbers f_2, f_3, f_4, f_5 , and f_6 .

Solution: The recurrence relation for the Fibonacci sequence tells us that we find successive terms by adding the previous two terms. Because the initial conditions tell us that $f_0 = 0$ and $f_1 = 1$, using the recurrence relation in the definition we find that

$$f_2 = f_1 + f_0 = 1 + 0 = 1,$$

$$f_3 = f_2 + f_1 = 1 + 1 = 2,$$

$$f_4 = f_3 + f_2 = 2 + 1 = 3,$$

$$f_5 = f_4 + f_3 = 3 + 2 = 5,$$

$$f_6 = f_5 + f_4 = 5 + 3 = 8.$$

Summations:

Summation of the terms of a sequence:

$$\sum_{j=m}^n a_j = a_m + a_{m+1} + \dots + a_n$$

The variable j is referred to as the index of summation, m is the lower limit and n is upper limit of the summation.

Example: Use summation notation to express the sum of the first 100 terms of the sequence $\{a_j\}$, where $a_j = 1/j$ for $j = 1, 2, 3, \dots$

Solution: The lower limit for the index of summation is 1, and the upper limit is 100.

We write this sum as $\sum_{j=1}^{100} \frac{1}{j}$.